Specialist Agent Record

Mei Qiong Xue | Assignment 6 | GitHub record updated 9 November 2026

A. Agent and version

Student / team: Mei Qiong Xue / individual assignment

Assignment / specialist: Assignment 6 / Emerging Technology Adoption Agent v1.1

Date: Analysis 2026-11-06; GitHub record updated 2026-11-09

GitHub repository: https://github.com/meix719/emerging-technology-agents

Branch: assignment-6-adoption

Tested implementation commit: e7345da9d7c4d4711e746cbeee26ea3e82d7b634

Frozen scaffold version: MASY1800_ET_Agent_Scaffold_v1_0; integrity check passed

B. Specialist purpose and context

Governing question: Which adoption posture is appropriate for this organization, application and time?

Emerging technology: Surgeon-controlled Mako robotic-arm assistance

Intended application: Primary elective total knee replacement for advanced symptomatic osteoarthritis

Organization / industry: Hypothetical teaching hospital A and community hospital B, England/NHS healthcare

Adoption posture: A: research-oriented fast follower. B: conservative, with concrete resource constraints.

Risk / consequence context: Patient harm is not reversible; a cancellable pilot does not remove clinical risk.

Decision / horizon: Current adoption posture; 12-month planning horizon, review within 90 days

C. Three-level finding

ET generally: Clinical use is established, but superior recovery is not established consistently.

ET for this application: Measure return to walking and daily activities against conventional total knee replacement.

ET in this organization: A can plan controlled learning under capability assumptions; B has capability gaps requiring deferral.

Bounded implication: A: conditional bounded experiment planning only. B: defer patient use.

Confidence / uncertainty: Moderate posture confidence; uncertain local benefit, safety applicability, cost and unverified readiness.

Reconsideration trigger: Relevant comparative evidence, verified safety/readiness, meaningful recovery gain and justified incremental cost

D. Evidence and testing

Most important evidence: S1-S2: NICE HTG743. S3-S4: RACER-Knee and trial team. S5: contrary early-benefit ROSA study. Full dated register is in the agent folder.

Primary test result: Conditional bounded experiment; no permission to enrol patients before clinical gates close.

Contrast - stable: The general technology evidence and intended application remain identical.

Contrast - changed: B defers due to training, support and measurement constraints.

Additional stress test: Aggressive A also defers when training and safety review are missing.

Weakness found: Actual pre-build conversation used vague evidence/comparability conditions; see preserved v0 output.

Revision: Define comparable evidence, label assumptions, require safety gates and direct recovery outcomes.

Remaining limitation: Three same-conversation tests do not establish reliability. Trial correction and real readiness remain unresolved.

E. Independent judgment and handoff

Credible for team consideration?: Yes as a bounded, reviewable prototype; not clinically validated or ready to authorize patient use.

Recommendation to team: Retain safety gates and conditional recommendations. If conditions cannot be represented downstream, use DEFER for immediate patient launch.

Organizational Adoption Agent handoff: Verify training, operational ownership, service support, rehabilitation capacity and implementation workflow.

AI / verification note: ChatGPT authored/tests the specialist using retrieved original-source materials; course validators, full JSON Schema checks and frozen-core integrity passed. Uploaded to GitHub on 2026-11-09. The implementation commit above preserves the tested agent; a subsequent documentation commit adds this completed record and reports.